OPEN SOURCE SECURITY MONITORING FOR HOSPITALS

Healthcare Threat Monitor

Technical reference: how every part works

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For	Hospital IT staff, security reviewers, auditors and developers
Project page	sea4real2000.github.io/Healthcare-threat-monitor
Source code	github.com/sea4real2000/Healthcare-threat-monitor

> This PDF is made from docs/technical_reference.md in the repository, which is always the latest version.

This document explains how every part of the platform works: what runs where, how a log becomes an incident, every API with examples, every setting, how the platform protects itself, and how to run it day to day. It is written for hospital IT staff, security reviewers, auditors and developers. You do not need to be a programmer to follow most of it; where code appears, the text around it says what it does. Words that may be unfamiliar are explained in the glossary at the end.

Other documents cover narrower ground and are linked where they matter:

Document	What it covers
hospital_install.md	Installing in a hospital, step by step, and how it has been tested
detection.md	The detection method in more depth
threat_model.md	How the platform itself could be attacked, and what limits each attack
public_demo.md	Running the public demonstration on free hosting
README.md	Starting points for developers
CHANGELOG.md	What changed in each version

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1. What it is, who it is for and how it is used

The problem it solves

Hospitals are attacked more than almost any other kind of organisation. Their systems cannot stop, so attackers expect a hospital under ransomware pressure to pay, and when an attack succeeds operations are cancelled and ambulances diverted. Patient records are also misused from inside: a member of staff opening the record of a celebrity, a neighbour or a colleague with no clinical reason.

The warning signs of both are usually already in the hospital's own logs, days before the damage. Large hospitals pay for a security operations centre to read them. Most hospitals, especially smaller ones and those with fewer resources, cannot, so nobody reads the logs in time. Healthcare Threat Monitor does that reading for free, on the hospital's own computer, and puts each warning in front of a person with the evidence needed to decide.

The platform in one page

Healthcare Threat Monitor watches the security logs a hospital already produces and warns early about three kinds of harm: ransomware and other attacks spreading through the network, staff opening patient records they have no reason to open, and medical devices behaving in ways they never should. A person reviews every warning and approves every response.

It works in seven steps:

Step	What happens
Collect	Logs arrive from Windows computers, routers, firewalls, servers, the patient record system and other sources. Each record is turned into one common format, called an event.
Detect	Twenty two written rules catch known signs of attack. Three learning models flag activity that is unusual for this hospital. A rule or model that finds something raises an alert.
Correlate	Alerts about the same machine or person within a short time join one incident, with one confidence score and a band: low, medium or high.
Review	An analyst sees the evidence, the attacker techniques involved and the reasons behind each alert, then confirms or dismisses the incident.
Respond	For a high incident, an analyst can approve cutting a machine off the network. Incidents can also be sent to a hospital's central security system.
Record	Every alert, decision and action is written to an audit log that shows if anyone changes a past entry.
Learn	A dismissal teaches the detectors that raised it, so the same false alarm is not raised again.

Everything runs on one computer inside the hospital. Nothing is sent to the project or to anyone else unless an administrator sets up forwarding to the hospital's own systems.

There are three ways to run it:

Way	For	What it is
Appliance	Hospitals	A ready made virtual machine (`.ova` for VirtualBox and VMware, `.vhdx` for Hyper-V) or a USB installer for a spare computer. Set up with clicks.
Hospital installation with Docker	Hospitals that already run Docker	`docker-compose.hospital.yml` on a Linux server. The appliance runs exactly this inside.
Demonstration	Anyone trying it	`docker compose up`, or hosted on Vercel and Supabase. Runs a simulated hospital with test attacks.

A hospital installation never runs the simulator, never accepts test attacks, has no public demo login, and needs sign in for every page.

Who uses it

Person	Account	What they use it for	Where in the console
The person who installs it, usually hospital IT	Administrator	Installs it, connects the log sources, sets up accounts, takes backups, changes settings	**Setup**, **Sources**, **Settings**
The person who watches it: an IT officer, a security analyst, or a shared duty in a small hospital	Analyst	Reviews incidents, confirms or dismisses them, approves isolating a machine	**Overview**, **Incidents**, **Assets**
The privacy or information governance officer	Analyst	Follows up incidents about patient records, which may lead to a conversation with a member of staff	**Incidents**
Leadership, auditors and regulators	Analyst, or an exported report	See that the hospital watches its systems, what was found and what was done	**Compliance**, **Audit**, the audit export
The hospital's central security system, where there is one	None: it receives forwarded incidents	Keeps incidents with alerts from its other tools	Set under **Settings**, SIEM forwarding

The platform does not need a dedicated security team. It is built so that one IT officer can watch a hospital, because only incidents worth a person's time reach the queue, and each one explains itself.

How it is used

The first day. Install the appliance, open its address, set up the administrator with the code on its screen, then follow **Setup**: load the departments, staff, computers and patients from spreadsheets, connect the patient record system, send the routers' and firewalls' logs to it, and install the Windows collector on the hospital's computers.

The first week. The learning period. The platform learns what is normal for each kind of log and reports only clear signs of attack. The person watching it checks **Sources** to see every log source arriving.

Every day. The person watching it opens **Overview** and **Incidents**, ideally at the start of each shift, as staff check other systems. Each incident in the queue shows what happened, on which machine or by whom, the evidence, and why each detector raised it. They confirm what is real and dismiss what is expected, and the platform learns from each dismissal.

When an incident is high. At least two independent sources agree something serious is happening, such as ransomware spreading. The analyst can approve isolating the machines it names, which the hospital's endpoint product carries out, and follows the hospital's own incident plan.

Every week or month. The administrator downloads a backup and keeps it off the computer. Leadership and auditors look at **Compliance** and **Audit**. When a new version is released, the administrator moves to it with a backup and a restore.

Four examples of the features working together

Ransomware on a ward computer. An email with a booby trapped attachment reaches a nurse (the email rule). Opening it makes Word start a script (the Office rule, from the Windows collector). The script copies passwords from memory (the credential dumping rule) and deletes the backup copies (the shadow copy rule). The firewall logs that computer scanning the network (the port scan rule, from the log receiver). Each alert joins one incident. Because the Windows log and the firewall log agree, it becomes high. The analyst sees the whole chain on one page, approves isolating the computer, and the connector carries it out. Every step is in the audit log, and the incident is forwarded to the hospital's SIEM.

A member of staff snooping. A records clerk in one department opens the record of a patient flagged as high profile, who is being treated in another (the flagged patient rule, from the patient record connection). The record access model also notices he opened many more records than they or their peers usually do. Only the record system can see this, so the incident stays medium: it is reviewed by a person, never acted on automatically. The privacy officer follows it up under the hospital's own policy.

A medical device behaving strangely. An infusion pump starts contacting an address on the internet (the device internet rule) and a host it has never contacted (the new peer rule), and its traffic model flags a burst far outside its normal pattern. The analyst sees the device, its department and what it contacted, and decides with the clinical engineering team what to do, because switching off a medical device is a clinical decision.

A false alarm, learned from. A new backup program on the server starts renaming many files every night, which looks like ransomware (the mass rename rule). The analyst checks with IT, dismisses the incident as expected, and the rule now needs one and a half times as many renames on that server before it alerts again. The change is listed on the Rules page and can be undone.

What it is not

- It is not antivirus and does not block anything by itself. It watches, explains and, with a person's approval, asks the hospital's endpoint product to isolate a machine.
- It does not read clinical content. From the patient record system it reads only the access log: who opened which record, and when.
- It does not replace the hospital's incident plan or its people. It gives them earlier warning and the evidence to act on.
- It does not send anything outside the hospital unless an administrator sets up forwarding to the hospital's own systems.

2. How the parts fit together

The parts

Part	Language	Folder	What it does
Console	TypeScript, Next.js	`web`	The web pages staff use, and the API every other part talks to.
Database	PostgreSQL 16	`supabase/migrations`	Holds events, alerts, incidents, settings, accounts and the audit log.
Background service	Python	`engine`	Detection, model training, reading records databases and FHIR servers, forwarding, retention.
Log receiver	Python	`engine/receiver`	Listens for syslog from routers, firewalls and servers on port 514.
Windows collector	PowerShell and C#	`web/collectors/windows`	Installed on Windows computers; sends their security events.
HTTPS server	Caddy	`docker/Caddyfile`	Serves the console over HTTPS with the installation's own certificate.
Backup service	Node.js	`web/scripts/backup_service.mjs`	Makes and restores backups when an administrator asks.
Appliance	Shell scripts	`appliance`	Builds the virtual machine and USB installer, and runs the screen.

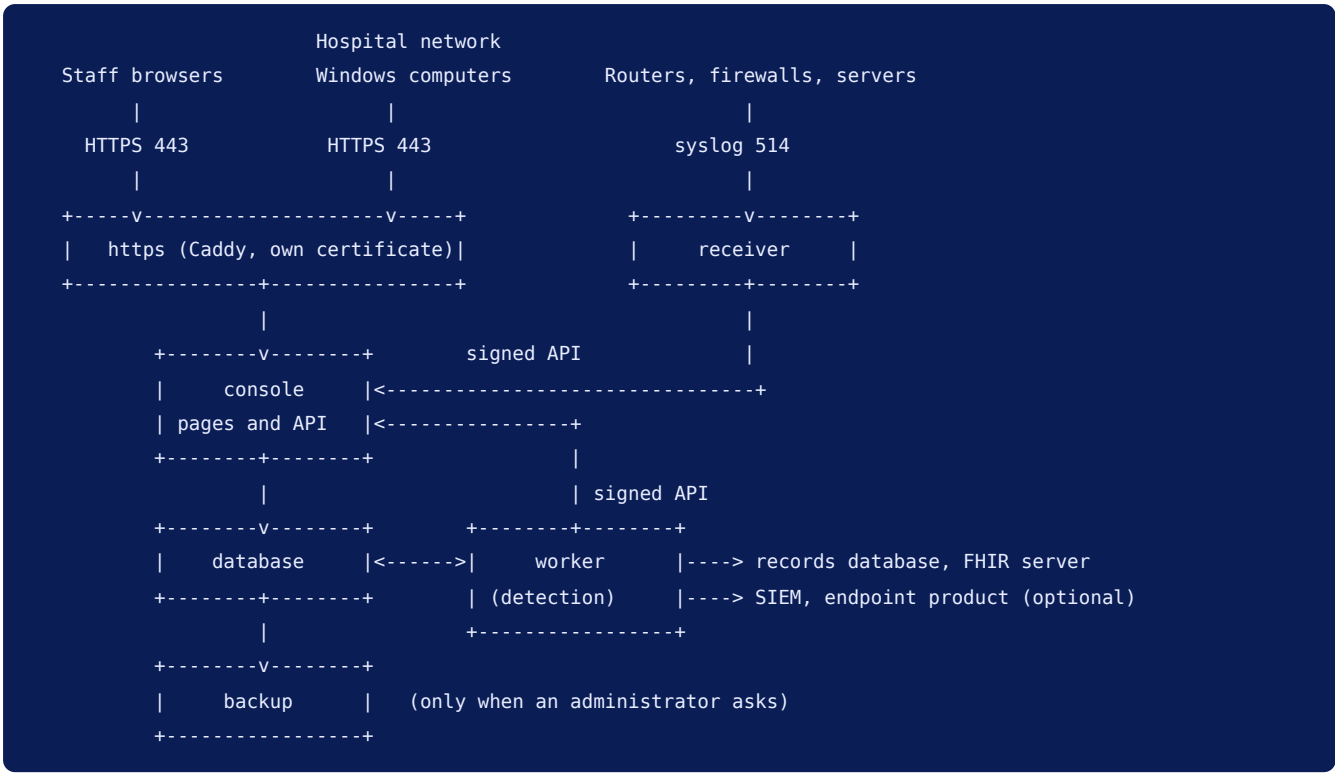
The hospital installation

`docker-compose.hospital.yml` starts these services. Each runs in its own container.

Service	Image	Role	Reachable from the network
`secrets`	tools	Makes the installation's secrets on the first start, then keeps them. Runs once and stops.	No
`db`	`postgres:16-alpine`	The database.	No
`setup`	tools	Applies database changes (migrations) and prepares the first visit. Runs once at each start and stops.	No
`console`	console	The web pages and the API.	Only through `https`
`https`	`caddy:2`	HTTPS in front of the console, on ports 80 and 443.	Yes: 80, 443
`worker`	worker	The background service.	No
`receiver`	worker	The log receiver, on port 514 (UDP and TCP).	Yes: 514

Service	Image	Role	Reachable from the network
`backup`	backup	Backups and restores.	No

The secrets live in a private Docker volume, readable only by these containers. The database port and the console's own port are never published, so the only ways in from the network are HTTPS and syslog.



The appliance

The appliance is a small Ubuntu 24.04 system that carries the images above and starts them. It has no remote sign in of any kind. Four of its own services matter:

Service	What it does
`htm-firstboot`	On the first start, grows the disk to its full size and loads the platform's software into Docker. Can be switched off at any point; it carries on next time.
`htm` (`htm-start`)	Starts the platform at the computer's current network address, so the certificate matches what staff type.
`htm-screen`	Shows the address, the setup code and the state on the computer's own screen. Press N for a fixed network address.
`htm-installer`	On the USB stick only: copies the platform onto a disk after the disk number and the word ERASE are typed.

3. Following one log through the platform

This is what happens when a nurse's computer runs a command that deletes backup copies, a common step just before ransomware locks files.

1. **The computer records it.** Windows writes a process creation event (number 4688) to its Security log.
2. **The collector sends it.** Within a minute, the Windows collector's scheduled task reads new events, keeps only the fields the platform reads, signs the batch with its group's secret and sends it over HTTPS to `POST /api/ingest`. It trusts only this installation's own certificate authority.
3. **The console checks and stores it.** The console checks the signature, the time stamp and that the group is switched on, turns each record into an event in the common format, matches it to the computer in the inventory, and stores it. A batch sent twice is stored once.
4. **The background service reads it.** Every few seconds (`engine.poll_seconds`), the worker reads new events and passes each through every rule. The rule `ransom_shadow_copy_delete` matches, with confidence 0.85, and raises an alert naming the MITRE ATT&CK technique T1490.
5. **Alerts become an incident.** The alert joins any incident about the same computer or person in the last 45 minutes, or opens a new one. The incident's confidence combines its alerts.
6. **The band decides what happens.** With only one log source agreeing, the incident cannot pass 0.79, so it is medium: it goes to the analyst queue and, if set up, is forwarded to the hospital's SIEM. If the firewall then logs the same computer scanning the network, two sources agree and the incident can become high.
7. **A person decides.** The analyst opens the incident, sees each alert's evidence and reasons, and either dismisses it (the detectors learn) or confirms it. For a high incident, the analyst can approve isolating the computer.
8. **The response is carried out.** The approval waits as a request. The endpoint connector reads it, asks the endpoint product to isolate the computer, and reports back. Only then is the computer marked isolated.
9. **Everything is recorded.** The incident opening, the analyst's decision, the approval and the isolation each add an entry to the audit log.

4. Collecting logs

The ten log sources

Every record is turned into an event with the same fields: when, which source, which machine, which person, what kind of action, its outcome and the details. These are the sources the platform reads:

Source	Typical origin	Examples of what it shows
`windows`	Windows collector	Sign ins, failed sign ins, new accounts, programs started, services installed, the security log being cleared
`firewall`	Log receiver, or a collector	Allowed and blocked connections, with addresses and ports

Source	Typical origin	Examples of what it shows
`vpn`	Log receiver	Remote sign ins and where they come from
`linux`	Log receiver	Server sign ins and failures over SSH
`ehr`	Records database, FHIR server or upload	Who opened which patient's record, and when
`edr`	Endpoint detection product	Files written, renamed and deleted, programs started
`dns`	DNS server	Domains looked up
`email`	Email security product	Messages and attachments delivered or blocked
`medical_device`	Network monitoring	Connections, hosts and packet rates per device
`cloud`	Cloud audit logs	Actions in cloud services

A source can be switched off on the Sources page. Records for a switched off source are refused with the reason `source_disabled`.

The Windows collector

An administrator creates a group of Windows computers under **Sources**, **Windows computers**, and downloads one installer file for it. Run on a computer as an administrator, it:

- copies the collector to `C:\ProgramData\HealthcareThreatMonitor`, which only SYSTEM and administrators can open;
- switches on process auditing, so programs started are recorded;
- adds a scheduled task that runs every `windows.interval_minutes`, and when the computer starts, as the SYSTEM account.

Each run reads new events of these kinds and sends them:

Log	Event	Meaning
Security	4624	Successful sign in
Security	4625	Failed sign in
Security	4672	Special privileges given at sign in
Security	4688	A program started
Security	4698	A scheduled task created
Security	4720	A user account created
Security	4732	A member added to a local security group
Security	4740	An account locked out
Security	1102	The security log cleared
System	7036	A service started or stopped

Log	Event	Meaning
System	7045	A new service installed

The position reached in each log is saved only after the platform confirms it stored the events, so nothing is lost when the network is down. On its very first run it sends only the last 24 hours.

The installer carries the installation's own certificate authority, and the collector trusts only that authority, so it cannot be tricked into sending events to another server. Each group has its own secret; switching a group off refuses its computers at once.

The log receiver

Routers, firewalls and servers send their logs by syslog to port 514 of the installation, over UDP or TCP. Both common syslog layouts are read (RFC 3164 and RFC 5424). Each message is then matched against these formats:

Format	Becomes
Cisco ASA	Firewall events
Fortinet FortiGate	Firewall and VPN events
pfSense and OPNsense filter log	Firewall events
Linux netfilter, including UFW and iptables	Firewall events
MikroTik	Firewall events
Linux sign in messages (sshd)	Linux events
Other key=value firewall logs	Firewall events, when the fields are recognisable

A device that sends for the first time waits on the Sources page until an administrator accepts it. Nothing from it is kept before that. Messages longer than `receiver.max_message_bytes` are cut; more than `receiver.sender_per_minute` messages a minute from one device are counted and dropped. Messages in a format the platform does not read are counted, so the Sources page shows what each device sends.

The receiver works out whether each address is inside the hospital or on the internet, and which machine it is from the inventory, so the network rules apply to router logs too.

The inventory

Detection needs to know the hospital: departments, staff and their roles, computers and devices, and patients with flags such as "staff member" or "high profile". The setup wizard under **Setup** loads these from spreadsheets, with a template for each (`GET /api/templates/{kind}`). Each row is checked, and a faulty file is refused with the row numbers to fix.

5. Patient record access

Watching who opens which patient record is one of the platform's main purposes. There are three ways to bring in the record system's access log. All three produce `ehr` events in the same format.

A. A read only database connection (recommended)

An administrator connects the record system's database under **Sources**, **Patient records**. PostgreSQL, MySQL and MariaDB are supported.

1. The administrator gives the address, database name, and a database account that can only read.
2. The platform lists the tables, guesses which columns hold the time, the staff member, the patient and the action, and shows a preview of the rows it would read.
3. Once saved, the background service reads new rows every `sources.poll_seconds`, at most `sources.batch_rows` at a time, and remembers where it stopped.

The connection refuses to write anything: every session is opened read only. The database password is sealed (encrypted) with `SOURCES_SECRET` before it is stored, and only the background service unseals it, when it connects.

B. A FHIR server

Many record systems offer FHIR, the international standard for exchanging health data. Its `AuditEvent` resource records who opened which record. The platform asks the server for new AuditEvents every poll. Four ways of signing in to the server are offered:

Way	When to use it
None	A test server only
Token	The vendor issued a fixed access token
Client secret	OAuth client credentials: a client ID and secret
Private key	SMART Backend Services. The console makes a key pair; the vendor registers its public half. Each request for an access token is signed with the private half (RS384), with the scope `system/AuditEvent.read` unless another is set.

Records are only ever read from the server's own address; a "next page" link pointing anywhere else is refused. Access tokens come from the token address the administrator gives, or the one the server publishes.

C. An uploaded report

Where neither is possible, an administrator uploads an access report exported from the record system as CSV or a spreadsheet. Unreadable rows are listed with the reason; rows uploaded twice are stored once.

6. Detection

This section is a summary. detection.md explains the method in more depth.

Rules

Rules are rows in the `rules` table and can be changed under **Rules** without touching code. Each has a kind:

- **match**: one event meets every condition;
- **threshold**: enough matching events for one machine or person within a time window;
- **new value**: a machine or person does something it has never done before.

The rules shipped with 0.9.0:

Area	Rule	Kind	ATT&CK	Confidence
Ransomware	Shadow copies or backup catalogue deleted	match	T1490	0.85
Ransomware	Credential dumping command	match	T1003.001	0.80
Ransomware	Mass file rename on one machine	threshold	T1486	0.80
Ransomware	Ransom note file created	match	T1486	0.75
Ransomware	Windows security log cleared	match	T1070.001	0.70
Ransomware	Office application started a script interpreter	match	T1204.002	0.60
Ransomware	Backup or shadow copy service stopped	match	T1489	0.50
Ransomware	Message judged malicious was still delivered	match	T1566	0.50
Ransomware	Risky attachment type reached an inbox	match	T1566.001	0.35
Access	Activity from a terminated account	match	T1078	0.80
Access	One machine signing in to many others	threshold	T1021	0.60
Access	Special privileges for a non administrator	match	T1078	0.55
Access	Repeated failed sign ins for one account	threshold	T1110	0.50
Access	Remote access from a new country	new value	T1078	0.35
Records	Flagged patient record opened from another department	match	T1213	0.45
Network	Lookup of a known malicious domain	match	T1071	0.80
Network	Internal port scan	threshold	T1046	0.60
Network	Large upload to the internet	match	T1048	0.60
Network	Lookup of a domain never seen before	new value	T1071	0.25

Area	Rule	Kind	ATT&CK	Confidence
Devices	Medical device sending a flood of packets	match	T1498	0.60
Devices	Medical device reaching for the internet	match	T1071	0.60
Devices	Medical device contacted an unfamiliar host	new value	T1046	0.60

Learning models

Three models each watch one kind of thing over short windows of time: staff record access (15 minutes), machine behaviour (1 minute) and medical device traffic (5 minutes). Each is an Isolation Forest, a method that learns what normal looks like without being shown attacks. A window raises an alert only when it is unusual overall and at least one measure is far outside the normal range for its peer group. Every alert lists the measures that drove it, with the value seen and the usual value. The models retrain every `engine.retrain_hours` on the last `engine.training_hours`.

Confidence and bands

Alerts from different detectors are treated as independent witnesses: if one is 60% sure and another 50%, the chance at least one is right is 1 minus (0.4 times 0.5), which is 80%. The incident's band follows from its confidence:

Band	Confidence	What happens
Low	below `detection.band_medium` (0.5)	Kept for tuning. Not queued, not forwarded.
Medium	from 0.5	Goes to the analyst queue and is forwarded to the SIEM.
High	from `detection.band_high` (0.8)	As medium, and an analyst may approve isolation.

An incident cannot be high unless at least two independent log sources agree (`detection.high_min_sources`). Until then it is capped at 0.79. A single faulty or faked sensor can therefore never lead to an isolation offer by itself.

The learning period

For the first `learning.days` (7) after each kind of log starts arriving, the platform learns what is normal in this hospital: which addresses, domains and countries are usual, and how much each person and machine normally does. During that time, clear signs of attack are still reported straight away; activity that is only unusual is reported once learning ends. **Sources** shows each kind of log's learning, with **Finish now** and **Learn again**.

Learning from analysts

When an analyst dismisses an incident as expected, each detector that contributed learns for that machine or person only: a new value joins its normal set, a threshold rises by `feedback.threshold_factor`, a match rule is silenced, and a model needs a larger difference before it alerts again. Every such change is listed on the Rules page and can be undone. A confirmed incident's activity is kept out of future training.

7. Response and forwarding

Isolation, with a person's approval

1. An analyst opens a high incident and approves isolation of a machine it names. Isolation is only offered on a high incident that names that machine and has not been dismissed. On the demonstration, the public demo visitor can also approve, within a rate limit.
2. The request waits in `response_actions` with the status `requested`.
3. A connector, a small program with the respond permission, reads waiting requests (`GET /api/response/actions`), acts on the endpoint product, and reports the result (`POST /api/response/actions/{id}/ack`) as `acknowledged` or `failed`.
4. Only an acknowledgement changes the machine's status to isolated. A failure leaves it unchanged and is shown on the incident.
5. Releasing a machine works the same way.

Out of the box, a hospital installation has no connector: approvals wait until one is set up. `engine/connectors/webhook_connector.py` posts each approved request to a webhook in the hospital's endpoint product, signed with `CONNECTOR_TARGET_KEY` and with an Authorization header from `CONNECTOR_TARGET_AUTH`. For a product with its own API, copy that file and replace its `deliver()` function. In the demonstration, the background service plays the endpoint itself (`response.simulated_connector`).

Forwarding to a SIEM

When an incident reaches medium or high, rises in band, is decided by an analyst, or has a machine isolated or released, a message is queued in `siem_outbox` and sent in up to two formats:

- **A JSON webhook** to `siem.webhook_url`, signed with HMAC SHA256 using `SIEM_WEBHOOK_KEY`, with the headers `x-htm-timestamp` and `x-htm-signature` (the same signing scheme as the API, below). Empty sends it to the built in receiver, so the forwarding can be watched working.
- **A CEF line** inside an RFC 5424 syslog frame, only when `siem.syslog_host` is set, to that host on `siem.syslog_port` over `siem.syslog_protocol`. Severity is 9 for high, 6 for medium and 3 for low.

A message that cannot be delivered is retried up to `siem.max_attempts` times, so nothing is lost when a destination is briefly down. Each message carries the newest audit log entry's hash at the time of sending. A copy kept outside the platform makes it possible to notice if the end of the audit log is ever cut off.

8. The API

The console's API is how every other part talks to it. Answers are JSON, except file downloads (the evaluation report, spreadsheet templates, the certificate, backups and pictures). A refusal from a JSON endpoint has this shape, with HTTP status codes used as usual (400 bad request, 401 not signed in or wrong signature, 403 not allowed, 404 not found, 409 conflict, 413 too large, 429 too many requests, 500 internal error, 503 not available):

```
{ "ok": false, "error": "signature_mismatch" }
```

Four ways a request proves who it is from

Way	Used by	How
Collector signature	Collectors, the receiver, the background service, connectors	HMAC SHA256 signature in three headers, below
Signed in session	Staff in a browser	The `htm_session` cookie, set at sign in. Requests that change something must also come from the installation's own pages.
Setup code	A new installation nobody has set up yet	The `X-Setup-Code` header, for restoring a backup only
SIEM key	The built in SIEM receiver	HMAC SHA256 with `SIEM_WEBHOOK_KEY`

Signing a request as a collector

Every collector has an id, such as `win_ab12cd34ef` or `hospital_service`, and a secret. The secret is never stored: it is worked out from the installation's master key each time, as HMAC SHA256 of `collector:{id}` with `INGEST_SIGNING_KEY`. Only collectors registered and switched on in the `collectors` table are accepted.

To sign a request, build this text, where the body hash is the SHA256 of the exact body sent, in lower case hexadecimal (an empty body for a GET):

```
v1:{timestamp}:{METHOD}:{path}:{sha256 of the body}
```

then send three headers:

Header	Value
`x-htm-key-id`	The collector id
`x-htm-timestamp`	The time now, in Unix seconds
`x-htm-signature`	HMAC SHA256 of the text above with the collector's secret, in hexadecimal

A request whose time stamp is more than `ingest.max_skew_seconds` (300) from the server's clock is refused, so a captured request cannot be replayed later. Requests are also limited per address by `limits.api_per_minute`.

A complete example in Python:

```
import hashlib, hmac, json, time, uuid, urllib.request

API = "https://10.0.0.5"
COLLECTOR_ID = "my_collector"
SECRET = "the secret printed for this collector"

def send(method, path, body=b""):
    timestamp = str(int(time.time()))
    text = f"v1:{timestamp}:{method}:{path}:{hashlib.sha256(body).hexdigest()}"
    signature = hmac.new(SECRET.encode(), text.encode(), hashlib.sha256).hexdigest()
    request = urllib.request.Request(API + path, data=body or None, method=method, headers
={
        "content-type": "application/json",
        "x-htm-key-id": COLLECTOR_ID, "x-htm-timestamp": timestamp, "x-htm-signature": sig
nature})
    return json.load(urllib.request.urlopen(request))

batch = {"batch_id": str(uuid.uuid4()), "records": [
    {"source": "firewall", "payload": {"id": "fw-000123", "ts": "2026-11-02T10:15:06Z", "a
ction": "deny",
     "src": "203.0.113.45", "dst": "10.20.0.1", "dpt": 22, "proto": "tcp"}}]}
print(send("POST", "/api/ingest", json.dumps(batch).encode()))
```

`engine/simulator/client.py` is a complete reference client, used by the background service itself. To print a collector's secret: `INGEST_SIGNING_KEY=... npm run --silent collector:key -- my_collector` in the `web` folder.

Endpoints for collectors and connectors

Method and path	Purpose
`POST /api/ingest`	Send a batch of log records
`POST /api/inventory`	Register or update departments, staff, machines and patients
`GET /api/runs/{run_id}`	What the database holds for one simulator run, per source (testing)
`GET /api/response/actions`	Isolation and release requests waiting to be carried out. Needs the respond permission.
`POST /api/response/actions/{id}/ack`	Report that a request was carried out, or failed. Needs the respond permission.

`POST /api/ingest` takes:

```
{
  "batch_id": "a new random UUID for each batch",
  "run_id": "optional, letters, digits and underscores",
  "records": [ { "source": "windows", "payload": { "...": "the record as the source wrote
it" } } ]
}
```

and answers:

```
{ "ok": true, "batch_id": "...", "replayed": false, "records": 2, "accepted": 2, "duplicat
e": 0, "rejected": 0, "rejects": [] }
```

A batch with the same `batch_id` sent again changes nothing and answers `"replayed": true`, so a collector can safely send again after a network failure. Each rejected record is listed with its position and reason (up to 20). Limits: `ingest.max_batch_records` records and `ingest.max_body_bytes` bytes per batch. A collector can be limited to certain sources (`collectors.allowed_sources`); a Windows group's collectors can only send `windows` records.

`GET /api/response/actions` answers with up to 100 waiting requests:

```
{ "ok": true, "actions": [ { "id": "42", "action": "isolate", "asset_id": "WS-ED-01", "ass
et_ip": "10.20.3.14",
  "asset_kind": "workstation", "incident_id": "117", "requested_by_name": "Ada Obi", "requ
ested_at": "..." } ] }
```

`POST /api/response/actions/{id}/ack` takes `{ "status": "acknowledged" }` or `{ "status": "failed", "note": "why, up to 200 characters" }`. A request that is no longer waiting answers 409 `action_not_waiting`.

Endpoints for signed in staff

Method and path	Who	Purpose
`GET /api/trust/root`	Any signed in account	Download the installation's certificate authority, `htm-root.crt`
`GET /api/collectors/windows/{id}`	Administrators	Download the installer for one group of Windows computers. It carries that group's secret.
`GET /api/audit/export`	Any signed in account	The whole audit log as JSON, for an outside auditor. Signed with `AUDIT_EXPORT_KEY` when set. `scripts/verify_audit_export.py` checks it independently.
`GET /api/evaluation/report`	Anyone	The latest detection evaluation report. It holds only results from the simulated hospital; a hospital installation, which runs no evaluation, has none.
`GET /api/templates/{kind}`	Anyone	A spreadsheet template for the setup wizard
`POST /api/backups`	Administrators	Ask for a backup: `{ "passphrase": "..." }`. Answers `{ "id": "..." }`.
`GET /api/backups/{id}/status`	Whoever holds the job id	How a backup or restore is going: `working`, `done` or `failed`, with the step
`GET /api/backups/{id}/file`	Administrators	Download a finished backup

Method and path	Who	Purpose
`POST /api/backups/upload`	Administrators, or the setup code on a new installation	Upload a backup file to restore, as the request body
`POST /api/backups/restore`	Administrators, or the setup code on a new installation	Restore an uploaded backup: `{ "upload": "...", "passphrase": "...", "confirm": "RESTORE" }`

The backup endpoints refuse a request whose `Origin` header is not the installation itself, so another website cannot trigger them through a signed in browser.

Other endpoints

Method and path	Purpose
`GET /api/health`	Health check for monitors. Says whether the database answers and which migrations are applied.
`GET /api/health?diagnose=1`	A fuller report, with the header `x-htm-diagnose-token` set to `DIAGNOSE_TOKEN`: table counts, tables without row level security, and whether `DATABASE_URL` and `INGEST_SIGNING_KEY` are set. It never shows a secret. Off unless `DIAGNOSE_TOKEN` is set.
`POST /api/siem/receive`	The built in SIEM receiver, signed with `SIEM_WEBHOOK_KEY`
`GET /api/media/{slot}`	A landing page picture (demonstration)

9. What is stored

Everything is in one PostgreSQL database. The main tables, grouped by purpose:

Purpose	Tables
The hospital	`departments`, `staff`, `assets` (computers and devices), `patients`
Logs	`events`, `ingest_batches`, `ingest_rejects`, `sources`, `collectors`, `log_senders`
Records connections	`record_sources`, `record_uploads`, `inventory_imports`
Detection	`rules`, `models`, `baseline_values`, `detector_exceptions`, `model_training_labels`, `learning_periods`
Incidents	`alerts`, `incidents`, `incident_entities`, `feedback`
Response	`response_actions`, `siem_outbox`, `siem_inbox`
Accounts	`users`, `sessions`, `rate_limits`
Audit	`audit_log`

Purpose	Tables
Configuration	`settings`, `copy` (every label and sentence), `site_images`, `compliance_controls`
Running state	`engine_state`, `evaluation_runs`, `scenario_runs`, `schema_migrations`

Retention. Events older than `retention.events_hours` (36) are removed. The evidence behind each alert is kept with the alert, so incidents keep their evidence after the events are gone. Alerts, incidents and the audit log are kept.

Database changes. Each file in `supabase/migrations` is safe to run again, and all are applied, in order, at every start of a hospital installation. Updating to a new version therefore needs no separate database step.

Lockdown. On a hosted demonstration with Supabase, the public database roles can read and write nothing; `supabase/tests/lockdown.sql` proves it, and CI checks that the test fails when a hole is opened.

10. Configuration

There are three layers, from rarely changed to often changed.

Secrets (environment variables)

On a hospital installation and the appliance, the database password, `INGEST_SIGNING_KEY`, `AUTH_SECRET`, `SIEM_WEBHOOK_KEY`, `AUDIT_EXPORT_KEY` and `SOURCES_SECRET` are made automatically on the first start, as 64 random characters each, and kept in the private secrets volume. Nobody needs to type them. The optional `DIAGNOSE_TOKEN` (at least 24 characters) and `RESEND_API_KEY` are only set by hand. On a hosted demonstration every value is set in the hosting provider, and each secret must be at least 32 characters.

Variable	Used by	Purpose
`DATABASE_URL`	Console, worker	Where the database is
`DATABASE_SSL`	Console	`require` or `disable`. Empty requires TLS except for local databases.
`INGEST_SIGNING_KEY`	Console	Every collector secret is worked out from it
`AUTH_SECRET`	Console	Seals each account's authenticator secret. Changing it means every account sets up its authenticator again.
`SIEM_WEBHOOK_KEY`	Console, worker	Signs forwarded incidents

Variable	Used by	Purpose
`SOURCES_SECRET`	Console, worker	Seals records database passwords and FHIR secrets and keys
`AUDIT_EXPORT_KEY`	Console	Signs audit log exports
`DIAGNOSE_TOKEN`	Console	Opens the detailed health report. At least 24 characters.
`RESEND_API_KEY`	Console	Sends invitation emails through Resend (optional)
`HTM_TRUST_URL`	Console	Where to read the installation's certificate authority for Windows installers
`HTM_BACKUP_DIR`	Console	The folder shared with the backup service
`HTM_API_URL`	Worker, connectors	The console's address
`HTM_COLLECTOR_ID`, `HTM_COLLECTOR_SECRET`	Worker	The collector the background service signs as
`HTM_CONNECTOR_ID`, `HTM_CONNECTOR_SECRET`, `CONNECTOR_TARGET_URL`, `CONNECTOR_TARGET_KEY`, `CONNECTOR_TARGET_AUTH`	Webhook connector	Carrying out isolations through a webhook
`HTM_DEVICE_DATASET`	Worker	Optional folder of CICIoMT2024 files for the demonstration
`HTM_HOSTNAME`	Hospital installation	The address staff type. The appliance sets it to its own address.

Settings

Every number the platform uses, such as thresholds, limits, intervals and bands, is a row in the `settings` table and can be changed by an administrator under **Settings**, with no restart. In the console, a missing or invalid setting is an error on purpose, so nothing silently falls back to a hidden default. The log receiver and forwarding keep safe built in values for a few settings, so logs are not lost if the database cannot be read for a moment. The appendix lists all 79 with what each does.

Wording, rules and pictures

To change	Where
Any label or sentence the console shows	**Settings**, **Labels and copy**
A detection rule, or add one	**Rules**
Landing page photographs	**Settings**, **Images**
The compliance mapping	Rows in `compliance_controls`

To change	Where
Colours, type and spacing	The token block at the top of `web/app/globals.css`

11. Accounts and sign in

Topic	How it works
Roles	**Administrator**: everything, including settings, accounts, sources and backups. **Analyst**: reads everything, reviews incidents, approves responses.
First administrator	On a new installation, every page leads to the first visit page, which needs the ten character setup code from the installation's own screen. The page closes for good once an administrator exists.
Other accounts	An administrator invites them under **Settings**, **Accounts**. The invitation link works for `auth.setup_link_hours`.
Passwords	At least `auth.password_min_length` (12) characters, not containing the name part of the email address. Stored as scrypt hashes (N = 2^15, r = 8, p = 3), never as the password.
Second factor	Every account must also give a six digit code from an authenticator app (TOTP). A code cannot be used twice. The authenticator secret is sealed with `AUTH_SECRET`.
Lockout	After `auth.max_failed_logins` (5) failures, the account locks for `auth.lock_minutes` (15). Each failure message is the same, so it never reveals which part was wrong.
Sessions	Random tokens, stored only as hashes, in a cookie that page scripts cannot read, sent only over HTTPS on a hospital installation, and not with forms posted from other sites. A session ends after `auth.session_hours` (12) or `auth.idle_minutes` (60) of no activity.
Rate limits	`limits.signin_per_minute` sign in attempts per address.

12. How the platform protects itself

threat_model.md goes through each kind of attacker in turn. The main protections are:

Protection	What it does
Signed collectors	Every record arriving is signed by a known, switched on collector, inside a five minute window. Secrets are worked out, never stored.
Least privilege	Only collectors with the respond permission can read or acknowledge isolation requests. A Windows group's secret can only send Windows events.
Human approval	No machine is ever isolated without a signed in person's approval, and never on one log source alone.
HTTPS everywhere	Caddy serves the console over HTTPS with the installation's own certificate authority; plain HTTP is sent to HTTPS.
Pinned trust	Windows collectors trust only the installation's own authority.

Protection	What it does
Content Security Policy	Pages may only run scripts carrying a fresh one time number (nonce), so injected script cannot run. Pages cannot be framed by other sites.
Cross site protection	Forms only accept posts from the installation's own pages; the backup endpoints check the `Origin` header.
Sealed secrets	Records database passwords, FHIR secrets and keys, and authenticator secrets are encrypted with AES 256 GCM before they are stored.
Tamper evident audit log	Each entry carries a SHA256 hash of itself and the entry before it. Changing or removing any past entry breaks the chain, and `audit_verify()` reports where. The database refuses updates and deletes on the log.
Locked backups	Backups made in the console are encrypted with AES 256 GCM under a key made from the administrator's passphrase by scrypt. A changed file is refused.
Minimal appliance	No SSH or other remote sign in. The maintenance account signs in only at the computer's own keyboard.
No secrets in the code	Every secret is made at installation or set in the host, never written in the code. CodeQL scans the code and ZAP attacks a running installation on every change.

Security problems should be reported privately, as SECURITY.md describes.

13. Backups, restores and updates

A backup holds everything: the database (incidents, sources, settings, accounts, connections, audit log), the secrets, and the installation's certificate authority. Restored onto a new installation, every account, connection and Windows computer keeps working.

Making one. An administrator opens **Settings**, **Backups**, chooses a passphrase of at least `backups.passphrase_min_length` characters and presses **Make the backup**, then **Download the backup**. The file waits on the installation for `backups.keep_hours` (24), then is removed.

The file. A `.htmbackup` file starts with `HTMBACKUP`, then holds the encryption settings and the contents encrypted with AES 256 GCM. The key is made from the passphrase with scrypt (N = 2^17, r = 8, p = 1). The passphrase is never stored: a lost passphrase cannot be recovered.

Restoring one. From **Settings**, **Backups**, or, on a new installation nobody has set up, from the first visit page with the setup code. The administrator chooses the file, types its passphrase and types `RESTORE`. The backup service then:

1. decrypts and checks the file; a wrong passphrase or a changed file is refused with nothing changed;
2. refuses a backup that holds database changes this version does not know, as one made by a newer version does;
3. replaces the database with the backup's, in one transaction, then applies any database changes the backup's version did not have;
4. ends every session, so everyone signs in again with the restored accounts;

5. restores the secrets, keeping the new installation's own database password;
6. restores the certificate authority, so a new certificate is made from it;
7. writes the restore to the audit log.

The console, background service and HTTPS server notice their keys or authority changed and start again by themselves within seconds.

Updating the appliance to a new version: make a backup on the old one, install the new appliance, restore the backup on its first visit page, then give the new one the old one's address. **Updating a Docker installation**: `git pull` and `docker compose up -d --build`; data and secrets are kept.

14. Running it day to day

Health

Check	How
Is it up?	`GET /api/health`, or the appliance's screen, which says **Running.**
Is detection running?	The console overview shows "Detection service running", from the background service's heartbeat.
Are logs arriving?	**Sources** shows each source, its last event, its learning period and silent sources (none for `console.source_silent_after_minutes`).
Is the log receiver running?	The Sources page warns when the receiver has not reported in for a minute.
Is the audit log intact?	**Audit**, or `select * from audit_verify()`

Logs of the platform itself

On a Docker installation, `docker compose -f docker-compose.hospital.yml logs SERVICE`, for example `logs worker` or `logs backup`. On the appliance, the same after the maintenance sign in, in `/opt/htm`. While the appliance is starting, it also writes each part's state to its serial port every 10 minutes, so a log can be taken from an installation that never finishes starting.

Common problems

Symptom	Likely cause and what to do
Browser warns about the certificate	Expected until hospital computers trust the installation's authority. Download it from **Sources**, **Windows computers**.
Browser says it cannot connect	The platform is not running yet. Wait for **Running.** on the screen, and type the address with `https://`.
A device's logs do not appear	Accept the device under **Sources**. Check it sends to port 514 of the installation's address.
A Windows group stops sending	The group is switched off, the computer cannot reach port 443, or the installation's address changed. Its `collector.log` says which.

Symptom	Likely cause and what to do
The address changed	Set a fixed address with N on the appliance's screen, or ask for a DHCP reservation. Windows computers set up at the old address must be set up again.
Backups page says the service is not running	Restart the installation.

15. How it is tested

Every change runs these checks on GitHub (`.github/workflows/ci.yml`), and each one fails the change if what it proves stops being true.

Check	What it proves
Console tests	Record formats, signing, passwords, codes, image checks, backup files. With coverage reporting and a floor.
Engine tests	Simulator, rules, models, receiver formats, FHIR, forwarding, connector. With coverage reporting and a floor.
Database and end to end	Every migration applies twice; visitor roles can read nothing; the audit log refuses and detects tampering; a short evaluation that fails if any attack is missed; the audit chain intact afterwards; detection checks on a fresh database.
One command demonstration	Starts from nothing and catches an attack.
Hospital installation	Starts from nothing over HTTPS and checks it is locked down; then a backup made with clicks, the installation removed, a new one started and the backup restored onto it in a real browser.
Dynamic analysis	ZAP scans and attacks a running installation, as a visitor and signed in, and fails on anything of medium risk or higher not accepted with a proven reason.
Windows collector	Installs the collector on a Windows computer as an administrator would, makes events, runs the scheduled task, and checks what a local stand in receiver, which checks every signature, receives. Also checks a wrong secret is refused and explained, and that uninstalling works.

CodeQL scans the JavaScript, TypeScript and Python code and the workflows on every change and every week, with its extended security queries (`.github/workflows/codeql.yml`). | Appliance scripts | Shell scripts checked with ShellCheck, the virtual machine description checked. |

For each release, `.github/workflows/release.yml` builds the appliance, starts it in a virtual machine, switches it off part way through its first start, starts it again, sets it up in a browser, sets and removes a fixed address, makes a backup, and installs from the USB stick with Secure Boot on, before the downloads are attached.

Both test suites report their coverage on every run, and each has a floor that only goes up as tests are added. The engine's figure also counts the end to end run, so it reflects the code a real detection run exercises.

16. Standards and outside services used

Standard or service	Used for
HL7 FHIR R4, AuditEvent	Reading patient record access from record systems
SMART Backend Services, OAuth 2.0	Signing in to FHIR servers without a person
Syslog, RFC 3164 and RFC 5424	Receiving router, firewall and server logs; sending CEF
CEF (Common Event Format)	Forwarding incidents to traditional SIEMs
Windows Event Log	What the Windows collector reads
MITRE ATT&CK	Naming the attacker technique behind each rule
HMAC SHA256	Signing every request between parts
AES 256 GCM, scrypt, HKDF	Sealing secrets, hashing passwords, locking backups
TOTP (RFC 6238)	Authenticator app codes
HIPAA Security Rule, NIST CSF 2.0, HHS 405(d)	The compliance mapping on the Compliance page
PostgreSQL, MySQL, MariaDB	Records databases the platform can read
Resend	Sending invitation emails (optional)
Supabase and Vercel	Hosting the public demonstration (optional)
scikit-learn	The Isolation Forest models
Next.js, React, PostgreSQL, Caddy, Docker, Ubuntu	What the platform is built on
GitHub Actions, CodeQL, Dependabot, ZAP	Testing, code scanning, dependency updates, attack scanning
CICIoMT2024 dataset, Synthea	Optional research data for the demonstration

17. Glossary

Word	Meaning
Alert	One detector finding one thing worth a look.
Appliance	A ready made computer system, here a virtual machine or USB installer, that runs the platform with no setup beyond clicks.
ATT&CK	MITRE's public catalogue of the techniques attackers use. Each rule names one.
Audit log	The platform's own record of everything that happened, chained so changes show.
Band	Low, medium or high: what an incident's confidence means for what happens next.
CEF	Common Event Format, a text layout many security systems read.
Certificate authority	The installation's own issuer of HTTPS certificates. Computers that trust it see no warning.

Word	Meaning
Collector	Anything that sends logs to the platform with a signature: the Windows collector, the receiver, the background service.
Confidence	How likely an alert or incident is to be a real problem, between 0 and 1.
Connector	A small program that carries out approved isolations in an endpoint product.
DHCP	How a network gives a computer its address automatically.
Endpoint product	Security software on computers that, among other things, can cut them off the network.
Event	One log record turned into the platform's common format.
FHIR	The international standard for exchanging health data.
HMAC	A way of signing a message with a shared secret, so the receiver can prove who sent it and that it was not changed.
Incident	Related alerts grouped into one problem for a person to review.
Isolation	Cutting one computer off the network so an attack cannot spread from it.
Isolation Forest	A learning method that spots unusual activity without being shown attacks.
Learning period	The first days after a kind of log arrives, while the platform learns what is normal.
Migration	A file that creates or changes the database's tables.
Nonce	A number used once. Here, a fresh one on every page decides which scripts may run.
SIEM	A central system where large organisations collect security alerts from many tools.
Syslog	The common way routers, firewalls and servers send their logs over the network.
TOTP	Time based one time passwords: the six digit codes an authenticator app shows.

18. Appendix: every setting

These are the settings as shipped with a hospital installation of 0.9.0. Administrators change them under **Settings**; changes take effect without a restart. On the demonstration, `engine.simulate` and `response.simulated_connector` are on.

Accounts and sign in

Setting	As shipped	What it does
`auth.idle_minutes`	60	A session ends after this long without activity.
`auth.lock_minutes`	15	How long a locked account stays locked.
`auth.max_failed_logins`	5	Failed sign ins before an account is locked.
`auth.password_min_length`	12	Shortest password accepted.
`auth.session_hours`	12	A session ends this long after sign in.
`auth.setup_link_hours`	48	How long an invitation link works.

Backups

Setting	As shipped	What it does
`backups.follow_minutes`	180	Minutes the backup page waits for a backup or restore to finish before saying it took too long.
`backups.keep_hours`	24	Hours a finished backup waits on the installation to be downloaded before it is removed. Uploaded backups that were never restored are removed after the same time.
`backups.max_upload_mb`	20480	Largest backup file, in MB, that can be uploaded to restore.
`backups.passphrase_min_length`	12	Shortest passphrase a backup can be locked with. The passphrase is never stored, so a lost one cannot be recovered.
`backups.restore_confirm_word`	RESTORE	The word to type to confirm a restore, which replaces everything on this installation.

Console display

Setting	As shipped	What it does
`console.assets_page_limit`	500	Most assets listed at once.
`console.audit_page_limit`	200	Most audit entries listed at once.
`console.incidents_page_limit`	100	Most incidents listed at once.
`console.overview_window_minutes`	60	Length of the window the overview and asset counts cover.
`console.recent_events_limit`	15	Rows in the latest events table.
`console.refresh_seconds`	15	How often the console asks for fresh data. Zero turns refresh off.
`console.source_silent_after_minutes`	10	A source with no event for this long is shown as silent.

Public demo

Setting	As shipped	What it does
`demo.auto_release_minutes`	20	While the demo is on, a machine stays isolated this long, then the background service releases it. Zero turns this off.
`demo.enabled`	false	Lets anyone use the public demo login. The demo visitor can launch scenarios and isolate or release machines, nothing else.
`demo.session_minutes`	60	A demo session ends this long after it starts.

Incident scoring

Setting	As shipped	What it does
`detection.band_high`	0.8	Confidence at or above this makes an incident eligible for isolation.
`detection.band_medium`	0.5	Confidence at or above this puts an incident in the analyst queue.
`detection.correlation_window_minutes`	45	Alerts on the same machine or person within this many minutes join one incident.
`detection.high_min_sources`	2	Independent log sources that must agree before an incident can be high.
`detection.incident_max_hours`	4	An incident stops taking new alerts this long after it opened.
`detection.single_source_cap`	0.79	Highest confidence an incident can reach while too few sources agree.

Email

Setting	As shipped	What it does
`email.from_address`	(empty)	Address invitations are sent from. Its domain must be verified with the email provider. Empty turns email off.
`email.from_name`	(empty)	Name shown as the sender. Empty uses the platform name.

Background service

Setting	As shipped	What it does
`engine.bootstrap_hours`	24	Hours of history the service generates on a new installation before first training.
`engine.poll_seconds`	2	How often the service looks for new events and requests.
`engine.retrain_hours`	6	How often the models retrain. Zero turns scheduled retraining off.
`engine.simulate`	false	Whether the service runs the hospital simulator.
`engine.training_hours`	24	Hours of recent activity each training run learns from.

Learning from analysts

Setting	As shipped	What it does
`feedback.global_after_entities`	3	A value dismissed as normal for this many entities becomes normal for all.

Setting	As shipped	What it does
`feedback.model_margin`	1.25	After a dismissal, a model alert for that entity must sit this multiple further from normal.
`feedback.normal_weight`	50	How heavily a window dismissed as normal counts in later training.
`feedback.threshold_factor`	1.5	After a dismissal, a threshold rule must see this multiple of the dismissed count for that entity.

FHIR connections

Setting	As shipped	What it does
`fhir.max_pages`	20	Most pages read from a FHIR server in one go.
`fhir.page_size`	100	AuditEvent entries asked for in each request to a FHIR server.

Hospital

Setting	As shipped	What it does
`hospital.name`	(empty)	The hospital's name, shown in the console. Set it in Setup.
`hospital.timezone`	UTC	The hospital's time zone. Times in its records with no zone of their own are read in this zone.

Ingestion

Setting	As shipped	What it does
`ingest.max_batch_records`	1000	Most records accepted in one batch.
`ingest.max_body_bytes`	4000000	Largest request body the ingestion API reads.
`ingest.max_skew_seconds`	300	How far a request timestamp may differ from server time.

Learning period

Setting	As shipped	What it does
`learning.days`	7	Days each kind of log is learned from, after its first record arrives, before unusual activity in it is reported. Clear signs of attack are reported straight away. Zero switches learning off.

Rate limits

Setting	As shipped	What it does
`limits.api_per_minute`	1200	Requests one address may make to the public API routes each minute.
`limits.demo_per_minute`	6	Demo logins, scenario launches and isolation requests one address may make each minute as the demo visitor.
`limits.signin_per_minute`	10	Sign in attempts one address may make each minute.

Landing page pictures

Setting	As shipped	What it does
`media.max_bytes`	3000000	Largest picture an administrator may upload, in bytes. Hosting limits a single upload to about 4 MB.

Anomaly models

Setting	As shipped	What it does
`model.alert_quantile`	0.995	A window must score above this share of training windows before it can raise an alert.
`model.confidence_max`	0.75	Highest confidence a model alert can carry.
`model.confidence_min`	0.35	Confidence of a model alert that only just crosses its threshold.
`model.full_confidence_z`	12	Deviations from normal at which a model alert reaches its highest confidence.
`model.min_feature_z`	3	At least one feature must be this many deviations from normal.
`model.min_group_windows`	200	A peer group with fewer training windows than this is judged against everyone.
`model.min_training_windows`	200	A model with fewer training windows than this stays silent.
`model.sample_size`	4096	Training windows each tree samples.
`model.trees`	200	Trees in each Isolation Forest.

Platform

Setting	As shipped	What it does
`platform.contact_email`	(empty)	Contact address shown on the landing page. Empty hides it.

Setting	As shipped	What it does
`platform.name`	Healthcare Threat Monitor	Working title shown in the console and the browser tab.
`platform.repository_url`	(empty)	Link to the public code repository, shown on the landing page.

Log receiver

Setting	As shipped	What it does
`receiver.batch_seconds`	5	How often the receiver passes the logs it has collected on for storage.
`receiver.max_message_bytes`	8192	Longest single log message the receiver keeps. Longer messages are cut to this length.
`receiver.sender_per_minute`	60000	Most log messages kept from one device in a minute. More than this are counted and dropped.

Response

Setting	As shipped	What it does
`response.connector_poll_seconds`	3	How often the simulated endpoint connector checks for isolation requests.
`response.simulated_connector`	false	Whether the background service acts as the endpoint. Switch off when a real connector is running.

Retention

Setting	As shipped	What it does
`retention.events_hours`	36	Events older than this are removed. Alert evidence is kept with the alert. Keep it above the training hours.

Test attacks (demo)

Setting	As shipped	What it does
`scenario.auto_interval_minutes`	0	Launch a scenario automatically this often. Zero turns it off.
`scenario.time_scale`	0.25	Speed of scenarios launched from the console. 1 is real time, 0.25 is four times faster.

SIEM forwarding

Setting	As shipped	What it does
`siem.max_attempts`	5	Delivery attempts before a message is marked failed.

Setting	As shipped	What it does
`siem.syslog_host`	(empty)	Syslog destination for CEF messages. Empty turns syslog off.
`siem.syslog_port`	514	Syslog destination port.
`siem.syslog_protocol`	tcp	Syslog transport: tcp or udp.
`siem.webhook_url`	(empty)	Where incident messages are posted. Empty sends them to the built in receiver.

Records databases

Setting	As shipped	What it does
`sources.batch_rows`	500	Most entries read from a records database in one go.
`sources.poll_seconds`	30	How often the background service reads new entries from each connected records database.

Uploads

Setting	As shipped	What it does
`uploads.max_bytes`	20000000	Largest report or spreadsheet an administrator may upload, in bytes. Vercel refuses uploads over about 4.5 MB whatever this says.

Windows collectors

Setting	As shipped	What it does
`windows.interval_minutes`	1	How often each Windows collector sends new events. Takes effect when a computer next downloads its settings.